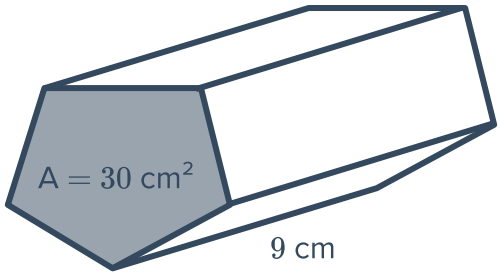
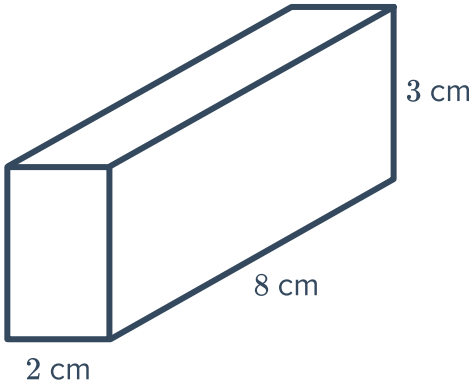
Volume of prisms

1 Find the volume of the following prisms. Round your answers to one decimal place where necessary.

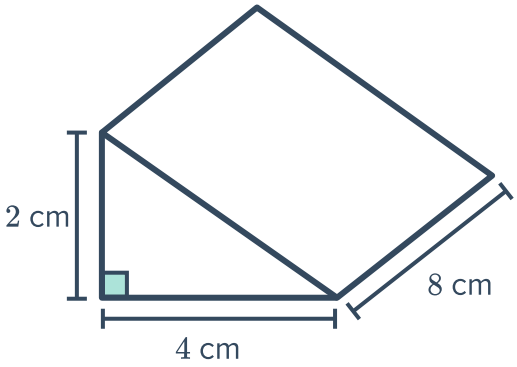
a



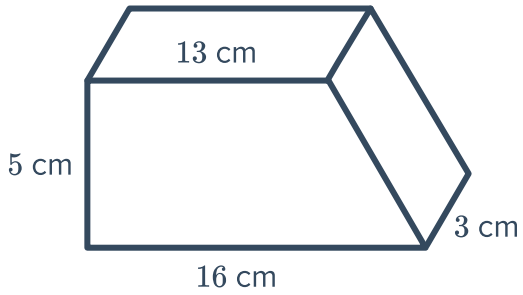
b



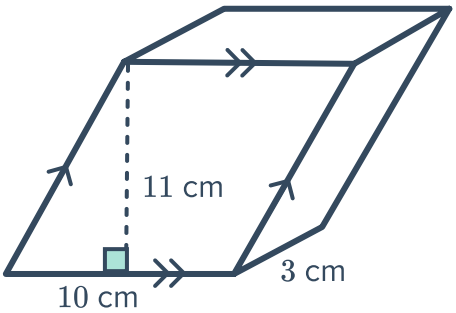
c



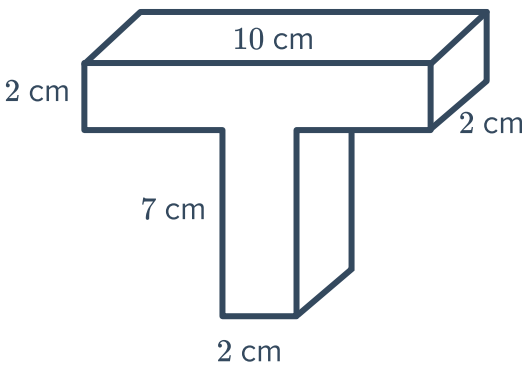
d



e

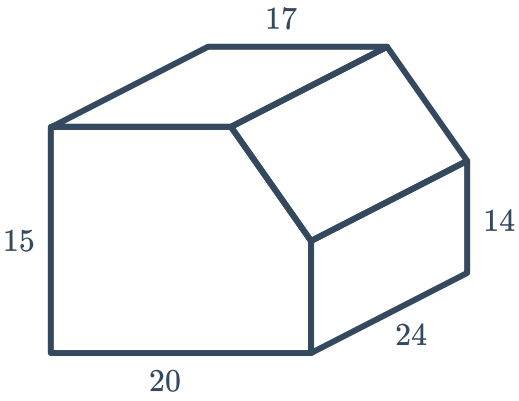


f

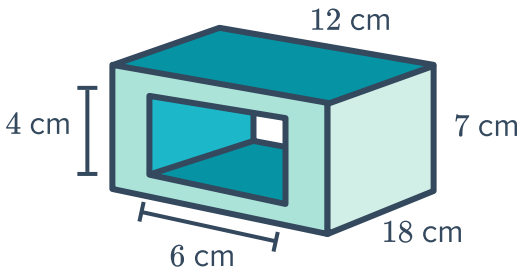


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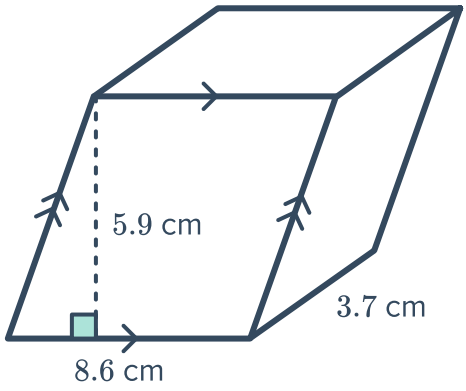
g



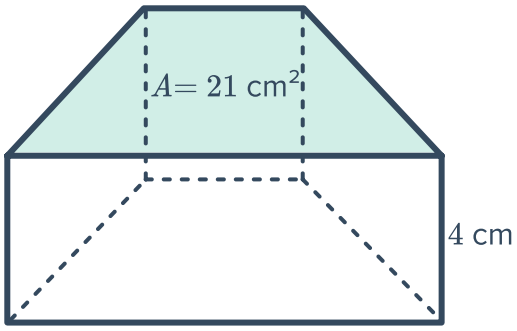
h



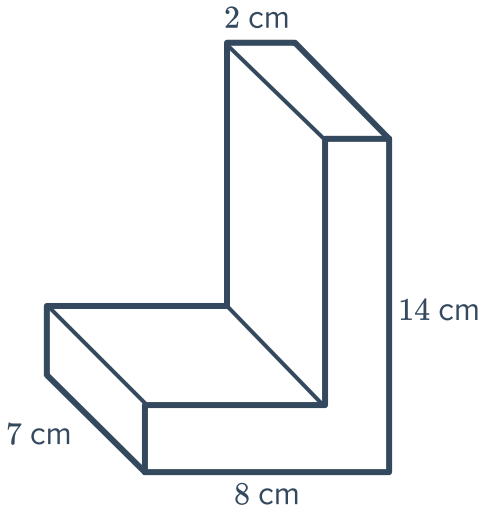
i



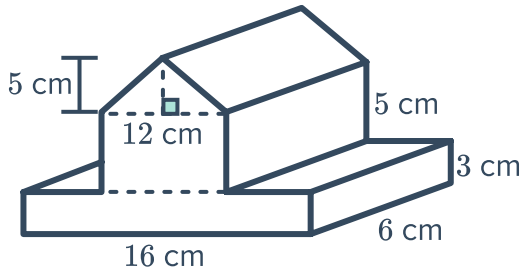
j



k

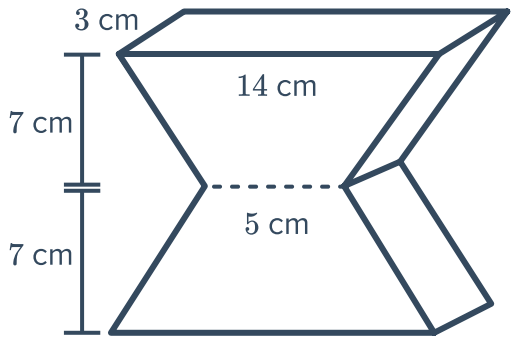


l

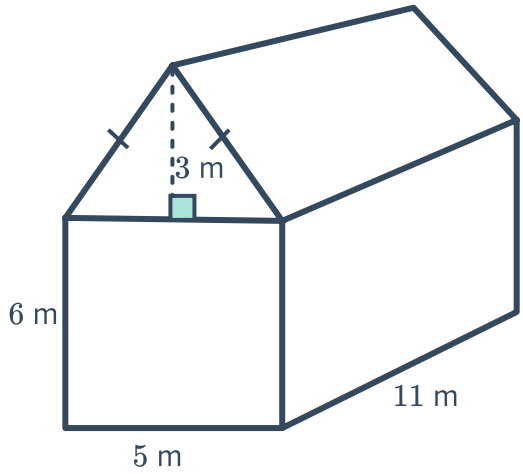


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m



n



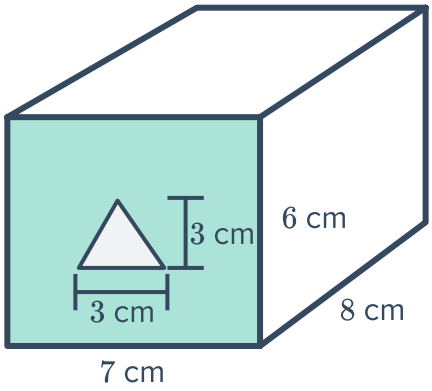
2 A cubic box has a volume of $34\,500\ \text{m}^3$.

a Find the side length of the cube, correct to four decimal places.

b Hence, find the volume of the largest ball that can fit inside the box, correct to the nearest whole number.

3 Find the cross-sectional area of a pentagonal prism whose volume is $470\ \text{mm}^2$ and height is $10\ \text{mm}$.

4 A triangular tunnel is made through a rectangular prism as shown in the figure. Find the volume of the solid formed.



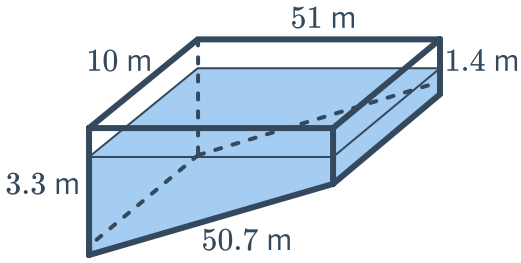
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5 A swimming pool has the shape of a trapezoidal prism as shown in the diagram:

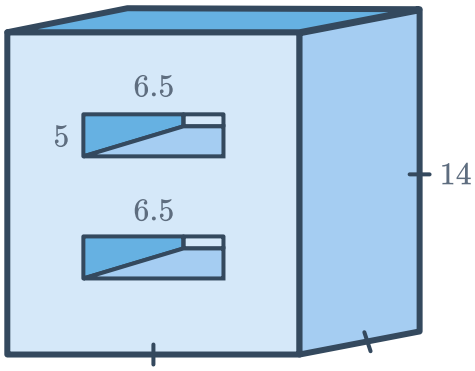
a Find the volume of the pool in cubic metres.

b If the pool is three-quarters full what is the volume of the non filled space of the pool.

c If the distance of the water level to the top of the pool is h m when it is is three-quarters full, then find h.



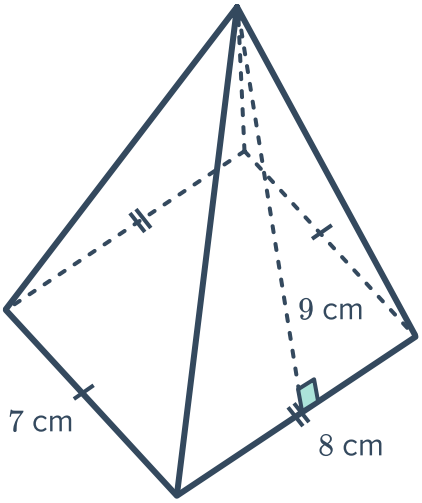
6 The larger prism has had two identical holes carved out of it, each of which is a rectangular prism. Find the volume of the remaining solid, correct to two decimal places. All measurements are in metres.



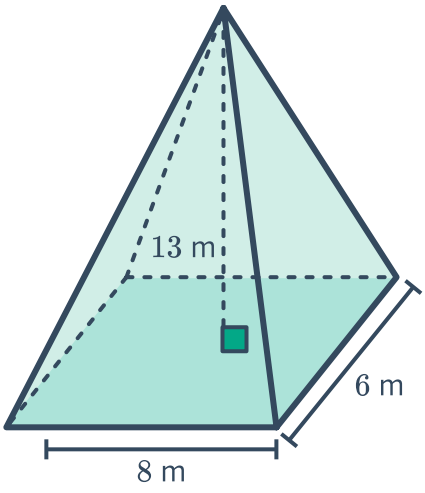
Volume of pyramids

7 Find the volume of the following pyramids, rounding your answers to two decimal places where necessary:

a

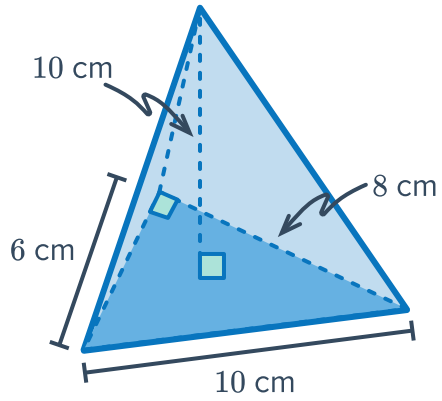


b

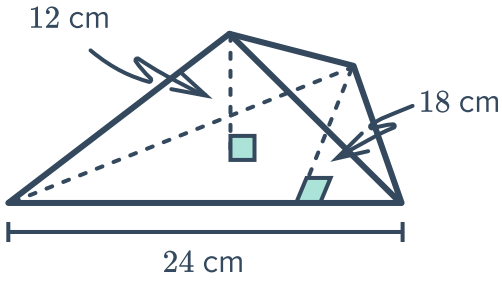


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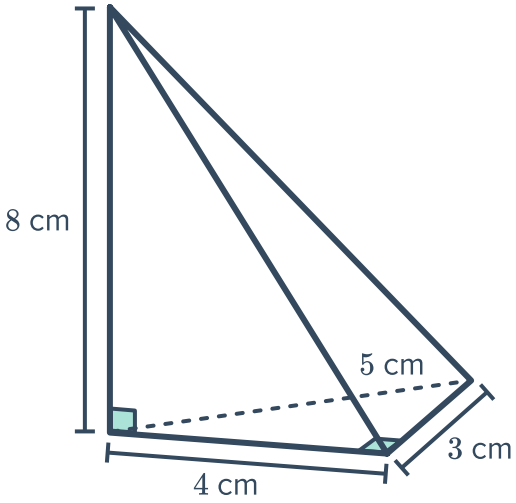
c



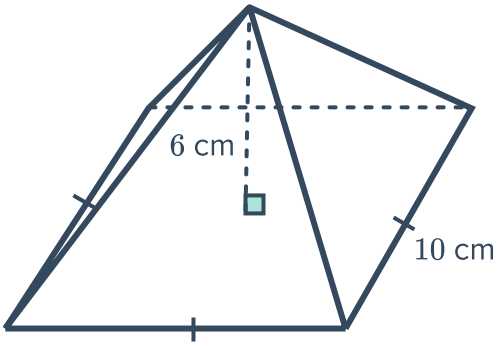
d



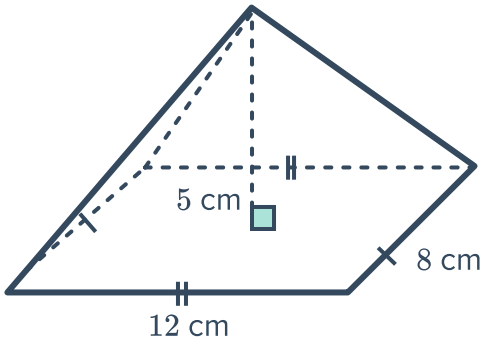
e



f



g



8 Find the volume of a pentagonal pyramid with base area of $90\ \text{cm}^2$ and height $8\ \text{cm}$.

9 A rectangular pyramid has a volume of $288\ \text{cm}^3$ cm. The base has a width of $12\ \text{cm}$ and length $6\ \text{cm}$. Find the height, h, of the pyramid.

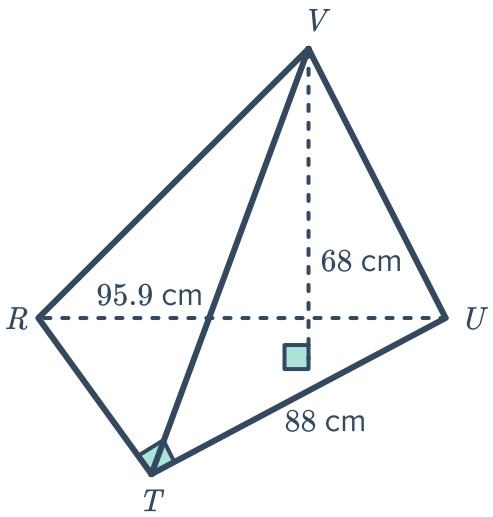
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10 In the triangular pyramid shown:

a Find the length of RT in centimetres, writing your answer correct to two decimal places.

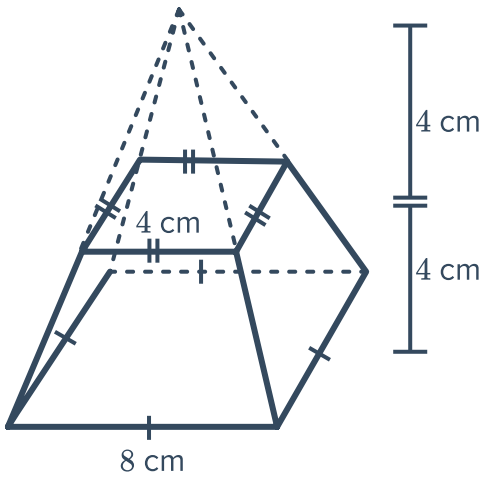
b Find the area of the base $\triangle RTU$.

c Hence, find the volume of the pyramid correct to two decimal places.



11 A rectangular pyramid has a volume of $432\ \text{cm}^3$. The height of the pyramid is 18 cm and the width of the base is 6 cm. Find the length of the base.

12 A small square pyramid of height 4 cm was removed from the top of a large square pyramid of height 8 cm forming the solid shown. Find the exact volume of the solid.

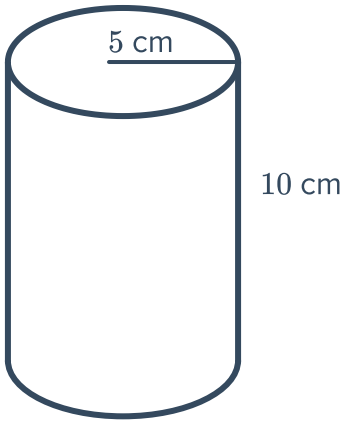


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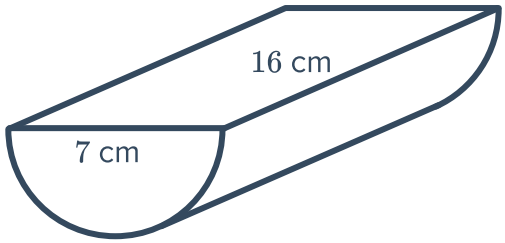
Volume of cylinders

13 Find the volume of the following solids, rounding your answers to one decimal place:

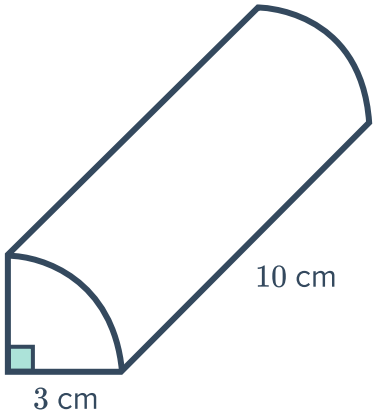
a



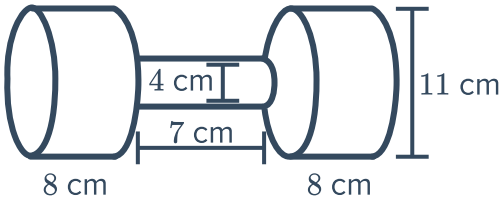
b



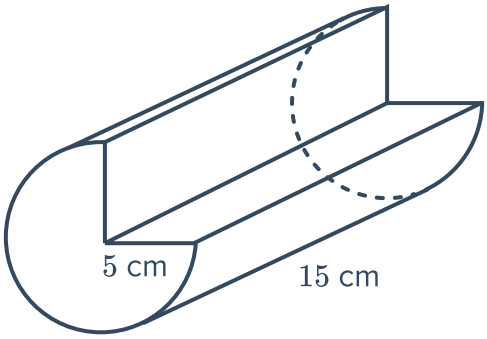
c



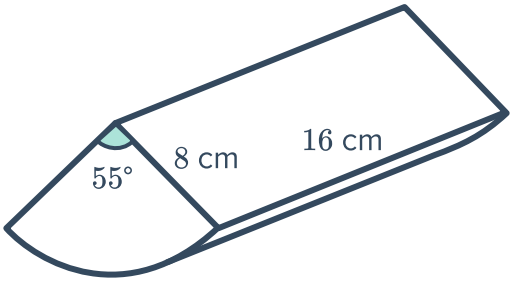
d



e

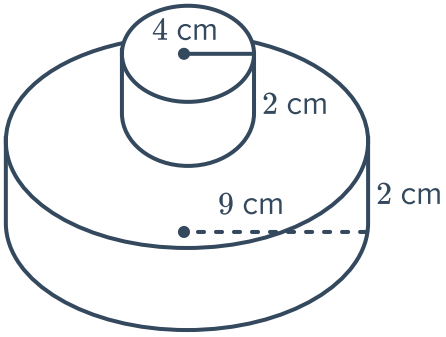


f



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g



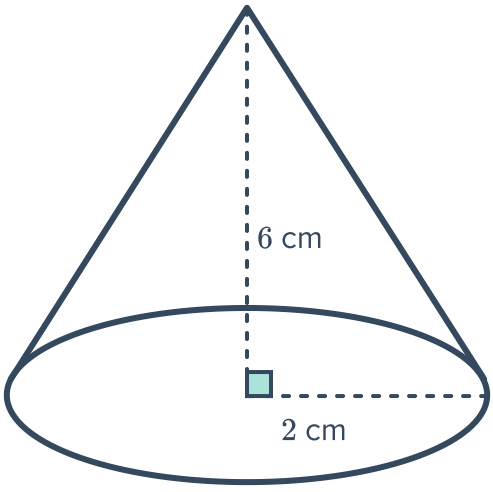
14 Find the volume of a cylinder with radius 7 cm and height 15 cm, correct to 2 decimal places.

15 If the radius of a cylinder is 8 cm and its height is 18 cm, find the amount of water it can hold in litres, correct to 2 decimal places.

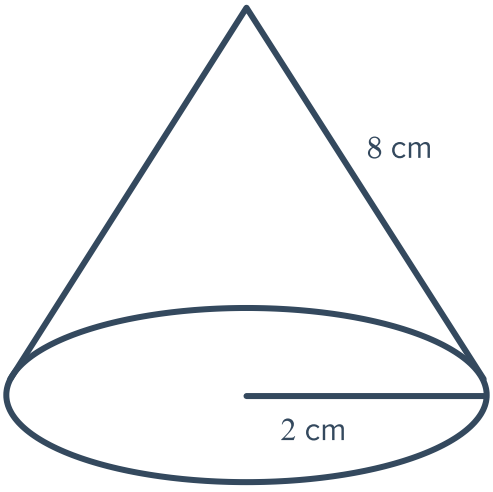
Volume of cones

16 Find the volume of the cones shown below. Round your answer to two decimal places.

a



b

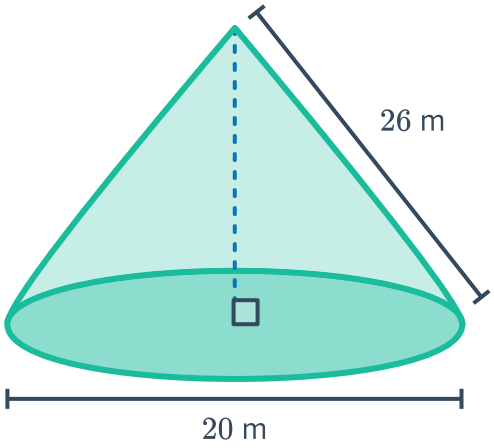


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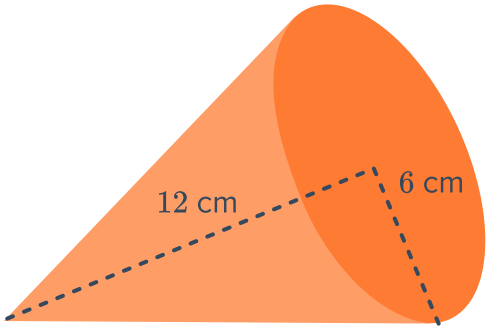
17 Consider the following cone:

a Find the perpendicular height.

b Find the volume. Round your answer to three significant figures.

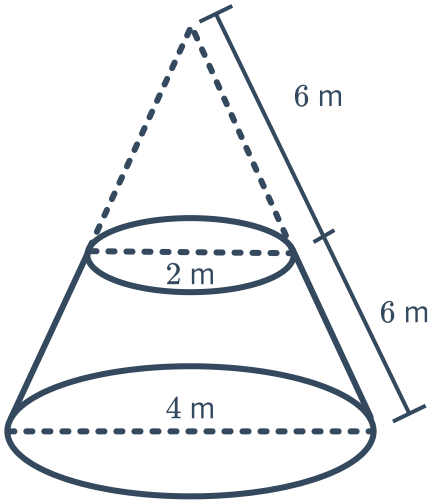


18 Find the volume of the cone shown. Round your answer to one decimal place.



19 A cone has a volume of $196\ \text{mm}^3$. If the height and radius of the cone are equal in length, find the radius, r. Round your answer to two decimal places.

20 The following podium was formed by sawing off the top of a cone. Find its volume, correct to two decimal places.

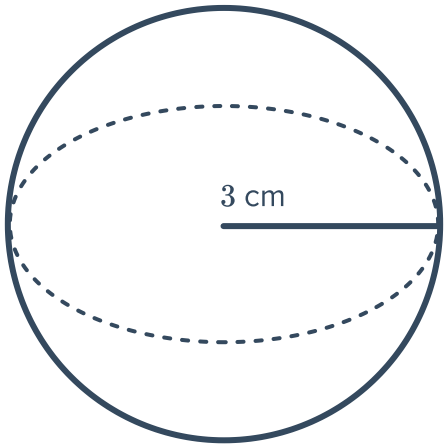


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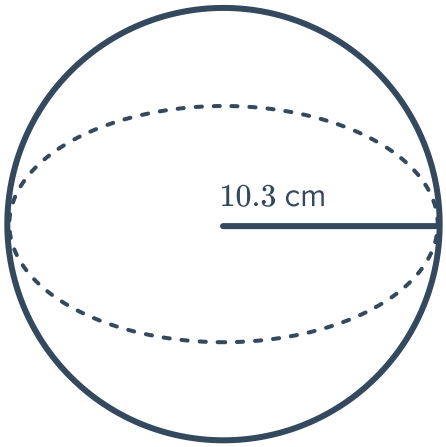
Volume of a sphere

21 Find the volume of the following spheres. Round your answers to two decimal places.

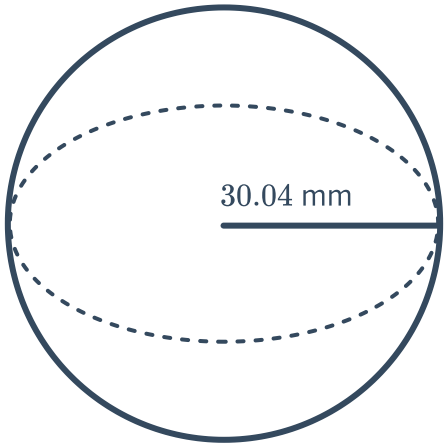
a



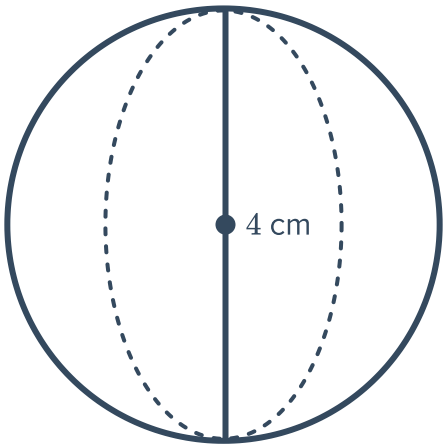
b



c



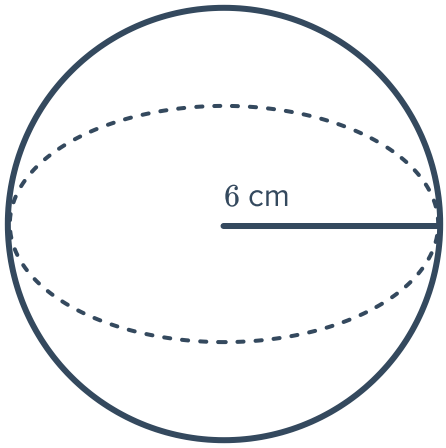
d



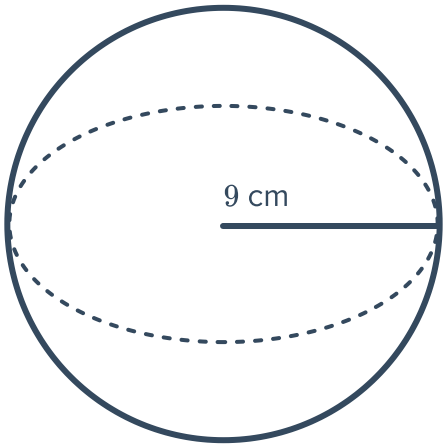
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22 Find the exact volume of the following spheres:

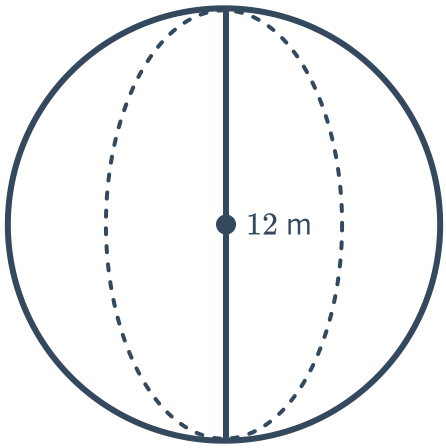
a



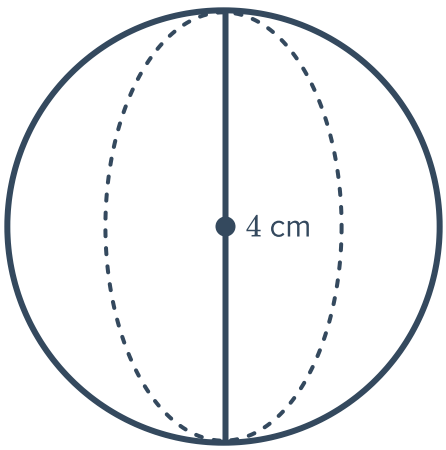
b



c



d



23 If the radius of a sphere doubles, what happens to its volume?

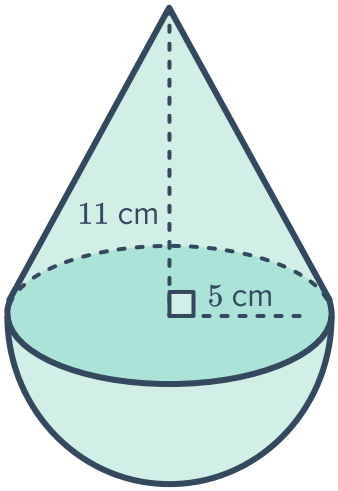
24 Find the radius, to two decimal places, of a sphere with volume of:

a $167\ \text{cm}^3$

b $\frac{343\pi}{3}\ \text{cm}^3$

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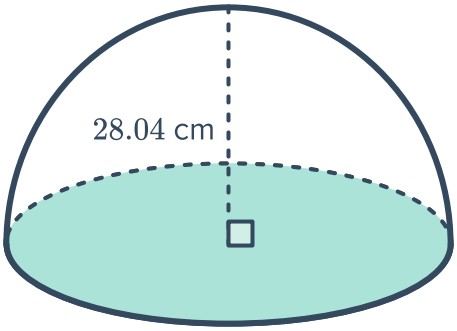
25 Find the volume of the composite figure shown.
Round your answer to two decimal places.



26 Consider the following hemisphere:

a Find its volume. Round your answer to three decimal places.

b Find its capacity in litres. Round your answer to the nearest litre.

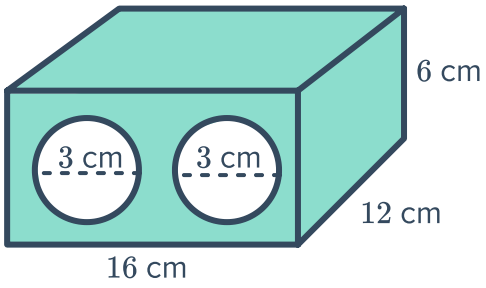


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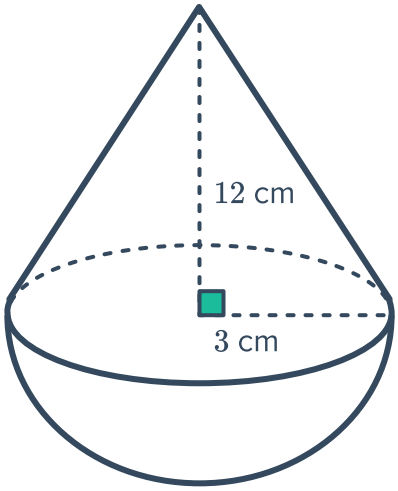
Applications

27 Find the volume of the solids correct to two decimal places:

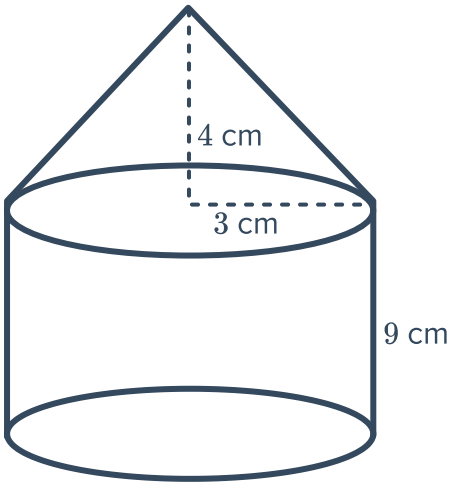
a



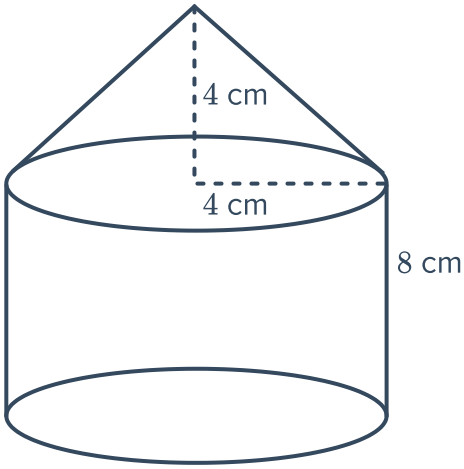
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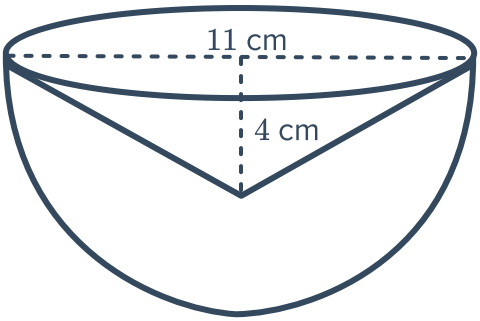
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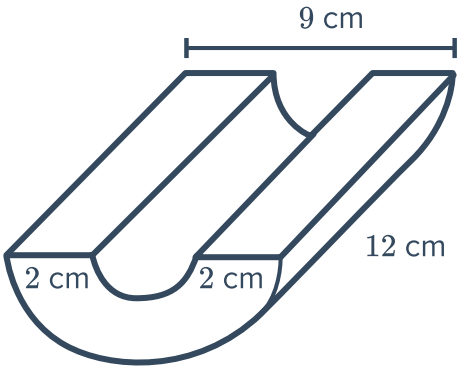
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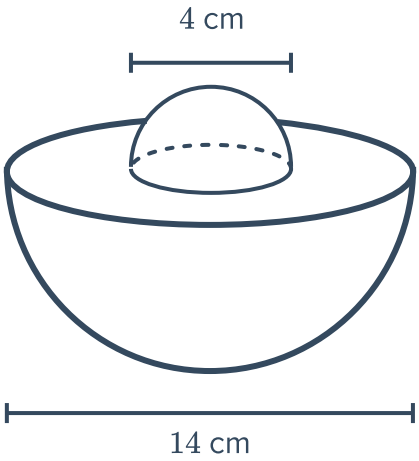


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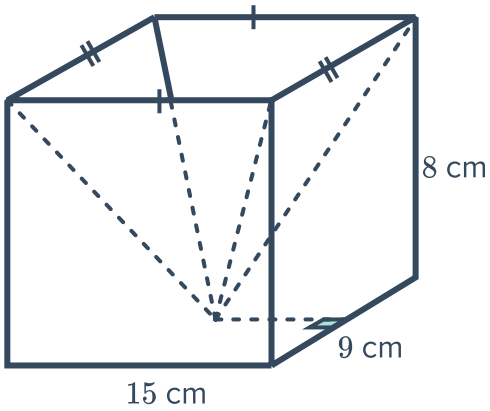


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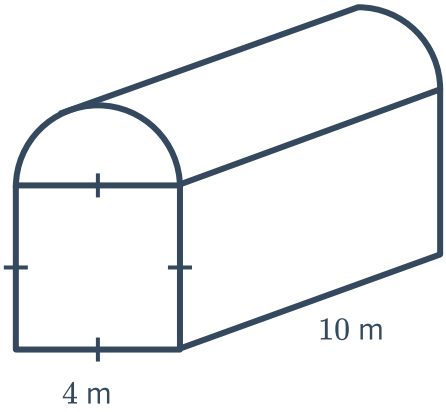
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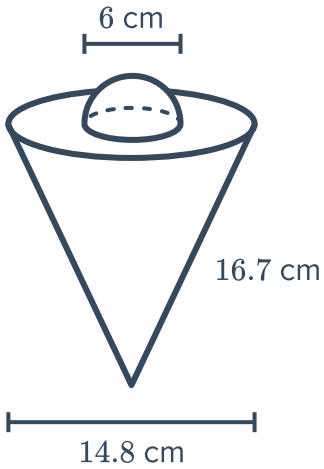
i



28 Consider the following solid:

a Find the perpendicular height of the cone, correct to one decimal place.

b Find the volume of the solid, correct to two decimal places.

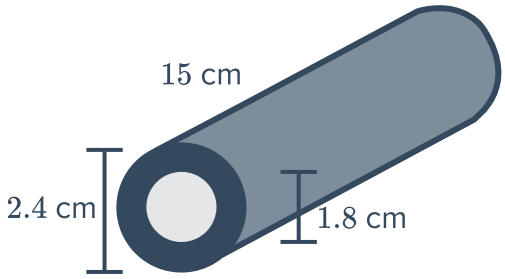


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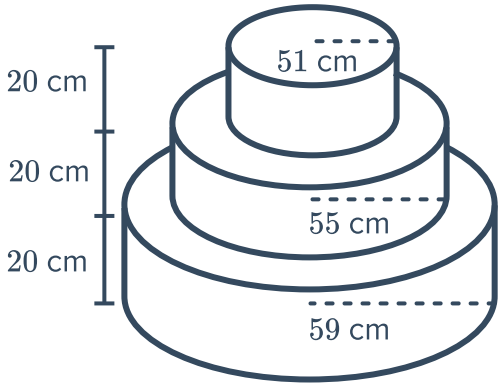
29 A hollow cylindrical pipe has the dimensions shown below:

a Calculate the volume of the pipe shown, correct to two decimal places.

b Calculate the weight of the pipe if $1\ \text{cm}^3$ of metal weighs $5.7\ \text{g}$, rounding your answer to one decimal place.



30 A wedding cake with three tiers is shown. The layers have radii of $51\ \text{cm}$, $55\ \text{cm}$ and $59\ \text{cm}$. If each layer is $20\ \text{cm}$ high, calculate the total volume of the cake in cubic metres.
Round your answer to two decimal places.



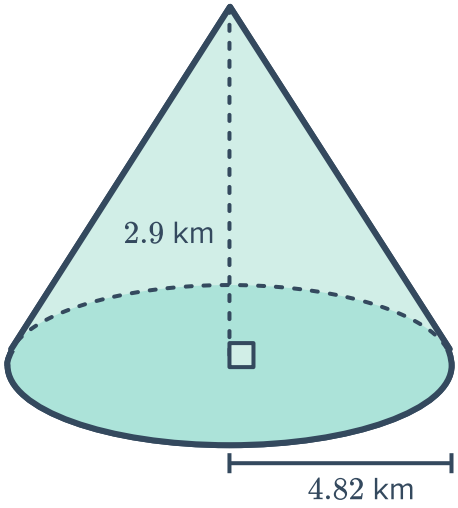
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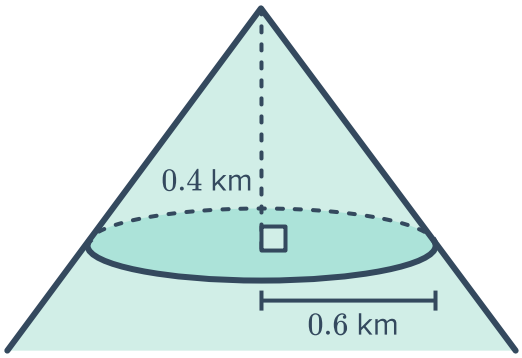
31 Before 1980, Mount St. Helens was a volcano approximately in the shape of the first cone below:

a What was the volume of the mountain, in cubic kilometres? Round your answer to two decimal places.

b The tip of the mountain was in the shape of the second cone shown. Find the volume of the tip in cubic kilometres. Round your answer to two decimal places.

c In 1980, Mount St. Helens erupted and the tip was destroyed. Find the volume of the remaining mountain, in cubic kilometres. Round your answer to two decimal places.

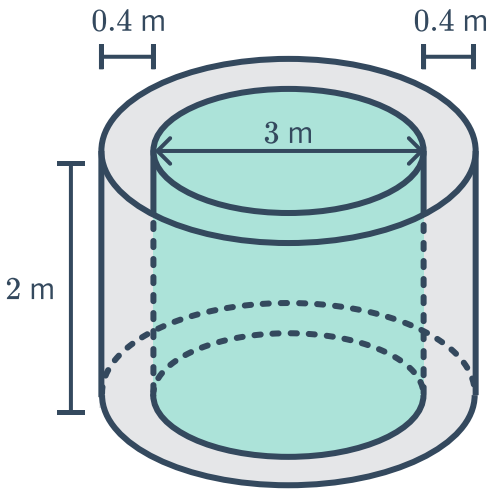




32 A cylindrical tank with diameter of $3\ \text{m}$ is placed in a $2\ \text{m}$ deep circular hole so that there is a gap of $40\ \text{cm}$ between the side of the tank and the hole. The top of the tank is level with the ground.

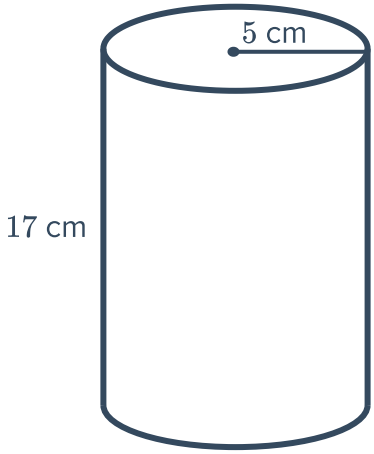
a What volume of dirt was removed to make the hole? Give your answer to the nearest metre cubed.

b What is the capacity of the tank to the nearest litre?



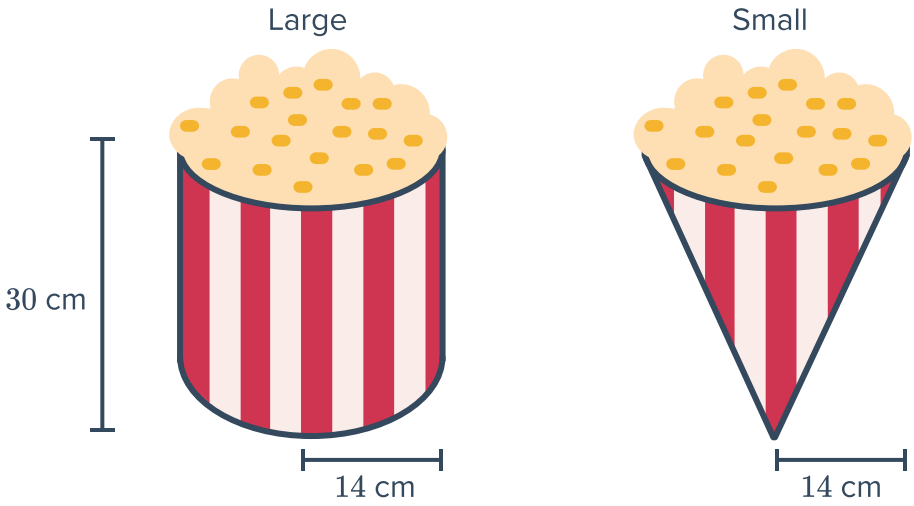
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33 Jack's mother told him to drink 3 large bottles of water each day. She gave him a cylindrical bottle with height 17 cm and radius 5 cm.



- **a** Find the volume of the bottle. Round your answer to two decimal places.
- **b** Assuming that he drinks 3 full bottles as his mother suggested, calculate the volume of water Jack drinks each day. Round your answer to two decimal places.
- **c** If Jack follows this drinking routine for a week, how many litres of water would he drink altogether? Round your answer to the nearest litre.

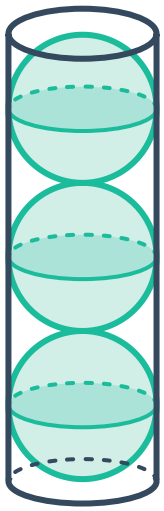
34 A theatre serves popcorn in two different containers. The Small size popcorn comes in a cone-shaped container, and the Large size popcorn comes in a cylindrical container as shown below. Both containers have a radius of 14 cm and a height of 30 cm.



- **a** How many small containers must be purchased in order to have the same amount of popcorn as in one large container?
- **b** Find the volume of the small container. Round your answer to the nearest cubic centimetre.
- **c** Find the volume of the large container. Round your answer to the nearest cubic centimetre.
- **d** The cost of a small container of popcorn is $\$1.30$ and the cost of a large container of popcorn is $\$3.50$. Which is the better buy?

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35 Three spheres of radius $4\ \text{cm}$ fit perfectly inside a cylindrical tube so that the height of the three spheres is equal to the height of the tube, and the width of each sphere equals the width of the tube.



a Find the total volume of the three spheres. Round your answer to one decimal place.

b Find the volume of the tube. Round your answer to one decimal place.

c Calculate the percentage of the space inside the tube that is not taken up by the spheres. Round your answer to the nearest whole number.

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